

Moon and has reached the Martian orbit but is far from mastering the aero engine technology. And no other technology has proven to be a greater strategic liability than inability to build an indigenous propulsion system.

Propulsion—whether fighter jet turbofans, submarine diesels, ship gas turbines, or tank engines—is the single most important technology that India cannot produce at scale. Despite 30-plus years of the Kaveri programme and the nuclear-powered INS Arihant, India will remain almost entirely import-dependent for propulsion across all services through 2034 and likely beyond.

This technology gap derailed the delivery timelines of the indigenous Light Combat Aircraft (LCA) Mk1A. In 2021, Hindustan Aeronautics Ltd (HAL) had contracted American General Electric for 99 F404-IN20 engines to power the 83 LCA Mk1A ordered by the IAF. The first engine arrived 14 months late. Only six engines have been delivered, underscoring how the American component has grounded the indigenous fighter jet ambitions.

“Jet engines remain a critical area where dependencies will remain for decades. Licensed manufacturing or co-development doesn’t mean fully independent. Dependency on original equipment manufacturers (OEMs) for critical parts will remain,” says Dinakar Peri, a fellow in the Security Studies programme at Carnegie India.

The Defence Research Development Organisation’s (DRDO’s) Gas Turbine Research Establishment has been working on development of the Kaveri engine since 1986. Four decades and thousands of crores later, the engine remains underpowered for fighter jets.

Ravi Kumar Gupta, former Scien-



THE PROPULSION GAP—FOUR DECADES AND COUNTING

Year	Programme	Status
1986	Kaveri engine programme launched	Still under development
2021	HAL contracts GE for 99 F404 engines	First engine arrives 14 months late
2026	Only six of 99 GE engines delivered	Critical delay. Supply chain dependence on US components has grounded indigenous fighter jet programme
2026	GTRE-Safran partnership for AMCA engine	In progress. France agrees to full technology transfer for the 5th generation Advanced Medium Combat Aircraft engine. Development begins
2035	AMCA engine-earliest delivery	If funded and prioritised—nearly 50 year after Kaveri project began



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tist G and Director, Public Interface. of DRDO, says the gaps have been allowed to be “formed, persist and widen by non-availability of adequate funds (eg, Kaveri Engine, semiconductor devices, advanced compact seekers) and/or bureaucratic hurdles, leading to creation of artificial gaps (like Titanium-related technologies)”.

“Ironically, annual foreign exchange outflow on procurements and MRO (maintenance, repair, and operations) of aircraft engines ranges between \$2 billion and tens of billions of dollars according to KPMG reports,” says Gupta.

GTRE is now partnering with France’s Safran to develop an engine for the 5th generation Advanced Medium Combat Aircraft (AMCA). France has agreed to full technology transfer. This means that India’s next generation fighter engine will arrive at the earliest by

the mid-2030s. Commander Milind Kulshreshtha (retd), a strategic analyst with an expertise in technologies related to C4I solutions, explains how shortage of key spares for gas turbine maintenance “adversely impacts warship deployment cycles, thereby the Fleet readiness levels.”

“As a solution, the Navy has put a multi-supplier system where propulsion components are purchased from diverse sources, including Indian vendors, for engine controls,” he says.

ELECTRONICS

Another consequential dependence is software. India has been seeking access to source codes for Rafale fighter jets. It is likely to be the first country besides France to produce these jets, but it will not have access to the source codes that form the electronic brain of the jet. The source code covers the Thales RBE2 AESA radar, the Modular Data Processing Unit, and